

 Presentation

Detection Engineering & Alert Triage

Improving SIEM capabilities, developing custom MITRE ATT&CK rules, and building a structured alert triage workflow.

Project Objectives



1. Log Validation

Verify that endpoint, network, and system logs are flowing into Wazuh properly, resolving any logging gaps across endpoints.



2. Custom Rules

Write custom XML detection rules in Wazuh mapped directly to MITRE ATT&CK techniques such as persistence and credential access.



3. Testing & Triage

Simulate attacks to test triggers, tune out high-volume false positives, and establish a repeatable incident triage workflow.

Lab Methodology



Phase 1: Ingestion

Validate baseline log telemetry and audit logging daemons across Ubuntu and Windows endpoints.



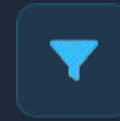
Phase 2: Rules

Develop custom Wazuh XML detection rules aligned with MITRE ATT&CK threat tactics and techniques.



Phase 3: Simulation

Execute Atomic Red Team scripts and manual attack triggers to test detection rule sensitivity.



Phase 4: Tuning

Apply white-list conditions to eliminate false positive noise and establish an alert triage process.

Log Source Validation

Gap Identified

Missing Auditd Service: System audit logging was not installed on the Linux endpoint. Because custom security rules depend on system audit logs, this missing source created a major telemetry blind spot.

Remediation Taken

Installed Auditd on Ubuntu, configured file/process monitoring rules, and linked log collection directly to the Wazuh agent. Successfully verified that Linux, Windows Sysmon, and Firewall logs flow cleanly.

Five Custom Detection Rules

Rule ID	MITRE ID	Technique Name	Target Threat Activity	Severity
100402	T1053.003	Cron Scheduled Job	Persistence via cron file modification	Level 8 (Med)
100405	T1543.002	Systemd Service	Persistence via systemd service creation	Level 9 (Med)
100407	T1059.004	Unix Shell Interpreter	Suspicious bash/python interactive execution	Level 10 (High)
100409	T1003.008	OS Credential Dumping	Memory inspection / dumping sensitive processes	Level 12 (Critical)
100410	T1552.001	Unsecured Credentials	Unauthorized access to shadow password file	Level 10 (High)

Attack Simulation

53,000+

High-Level Alerts Generated

Simulation & Alert Flooding

- 🔧 **Execution:** Executed attack simulations using Atomic Red Team scripts and manual terminal commands.
- ✓ **Rule Triggers:** All 5 custom rules triggered successfully upon detecting adversary actions.
- 🔊 **Noise Spike:** Broad detection parameters in Rule 100409 caused an overwhelming flood of over 53,000 duplicate events.

Noise Tuning & Optimization



Rule 100407

Terminal Filtering: Restricted shell execution alerts to interactive terminal sessions (pts/X, ttyX), excluding non-interactive background tasks.



Rule 100410

Process Exceptions: Added exemptions for safe administrative binaries like sudo, su, sshd, and passwd.



Rule 100409

Binary Exclusion: Whitelisted harmless debugging processes (e.g., strace, gdb), stopping massive duplicate alert generation.

Alert Triage & Incident Handling

Standard Triage Process

Established a step-by-step workflow for analysts to validate alert severity, collect forensic evidence, investigate host actions, and determine escalation path.

Triage Breakdown (9 Alerts)

-  **7 True Positives:** Real simulated attack actions successfully captured by custom rules.
-  **2 False Positives:** Harmless administrative actions correctly identified and closed out.

Key Findings

-  **Logging Precedes Detection:** Custom rules cannot alert on attacks if telemetry is missing at the endpoint source (e.g., missing Auditd daemon).
-  **Tuning Eliminates Fatigue:** Raw, un-tuned rules cause severe alert fatigue; continuous tuning is mandatory for operational SOC balance.
-  **Auditd & SIEM Synchronization:** Wazuh rule evaluation requires exact alignment with Linux Auditd keys to decode events accurately.
-  **Structured Workflows Drive Consistency:** A repeatable triage workflow ensures uniform investigation quality across all analyst tiers.

Recommendations



1. Baseline Activity

Continuously test detection rules against normal user workflows to catch and filter new false positives early.



2. Maintain Configs

Keep Auditd rules and Wazuh XML definitions synchronized whenever system configurations or OS updates occur.



3. MITRE Mapping

Expand custom rules to cover additional MITRE ATT&CK tactics like Defense Evasion and Lateral Movement.

Thank you! Any questions?

SOC Detection Engineering & Alert Triage



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